

Linear Algebra Overview Part 1

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1 Introduction

What is linear algebra and why do we care? Well, it's where stuff's linear, and we care because linear stuff is easier. Now in all seriousness linear algebra is the study of vector spaces. And we are interested in vector spaces because vector spaces are a type of mathematical construct that has the capability to model systems, linear transforms, and many more aspects in physics. As engineers, vector spaces are the mathematical construct we typically use to create our mathematical models from. This is because Newtonian physics, which is the primary area in which mechanical engineering and its sub fields are done, is built off the assumptions of linear space.

To begin we are going to discuss what it even means to be linear and what linear combinations are. The basic idea is right there in the name, something is linear if it is line-like, specifically it holds the properties of a what you would think of when thinking about a line. Now that's where it gets complicated, what is a line. There are a few ways we could look at this question whether we are considering Euclidean/Plain Geometry where our space has curvature zero, or if we were looking at Non-euclidean Geometry, or maybe Riemannian Geometry, or even possibly Manifolds.

(I bring this up as gravitation problems can be modeled using Manifolds which is very applicable to aerospace engineering, but is outside the scope of these lectures and notes. I do encourage the reader to look into this. I also offer a warning as to truly dig into manifold geometry requires foundations in both topology and analysis. This should not discourage you but rather inform you of where to look for clarification when you get confused.)

Returning to Linear Algebra, we will consider the case of linearity in the algebraic sense because we are doing linear algebra. This means that we have additive combinations with scalars as weights. Not dissimilar to how grades are calculated and in fact that will be one of our examples!

2 Definitions and Theorems

Linear Equations

Let $a_i \in \mathbb{R}, \forall 0 \leq i \leq n$ and let $b \in \mathbb{R}$. Further assume $x_i \in \mathbb{R}$ are in the domain. Then a linear equations is an equation of the form

$$a_0x_0 + a_1x_1 + \dots + a_nx_n = b$$

Where all a_i, b are constant.

Linear Systems of Equations

Let $a_{ij}, b_j \in \mathbb{R}, \forall 0 \leq i \leq n, \forall 0 \leq j \leq m$ and assume $x_i \in \mathbb{R}$ are in the domain. Then a linear system of equations is a set of linear equations of the form

$$a_{00}x_0 + a_{01}x_1 + \dots + a_{0n}x_n = b_0$$

$$a_{10}x_0 + a_{11}x_1 + \dots + a_{1n}x_n = b_1$$

...

$$a_{m0}x_0 + a_{m1}x_1 + \dots + a_{mn}x_n = b_m$$

Matrices

A Matrix is a rectangular array of entry's of mathematical objects. Suppose $n, m \in \mathbb{N}$ are natural numbers, then a matrix M is called an $n \times m$ matrix and is of the form

$$\begin{bmatrix} a_{11} & a_{12} & \dots & a_{1m} \\ a_{21} & a_{22} & \dots & a_{2m} \\ \vdots & \vdots & & \vdots \\ a_{n1} & a_{n2} & \dots & a_{nm} \end{bmatrix}$$

where $a_{ij} \in \mathbb{R}$

Note: formally, the entry's of a matrix need to be elements of a commutative ring, but this is extra for those interested in more rigor.

This definition assumes the ring is the real numbers $\mathbb{R}$, but the complex numbers $\mathbb{C}$ work as well. Another example could be the set of continuous functions.

Linear Combinations

Let V be a real vector space and $v_0, v_1, \dots, v_n$ be vectors and $a_0, a_1, \dots, a_n$ be scalars then a linear combination is of the form

$$a_0 v_0 + a_1 v_1 + \dots + a_n v_n$$

3 Examples

Lets start our examples by looking at some linear and nonlinear equations. Consider the following examples where x, y, z are variables.

A $x + \pi = e^y$

B $42x + 67y + 73z = \ln(\pi)$

C $\sin(4)x = y$

D $xy + z = 5$

E $4\pi = (21x)^2 - \frac{17}{59}y$

Equation A has the y in an exponential so this is nonlinear. Equation B has some funny numbers as coefficients, but it is still linear. And bonus point to whom ever knows the significance of why 73 is cool! Equation C is kinda deceiving but remember $\sin(a)$ where a is constant is itself a constant so C is linear. Equation D is tricky because x and y are each others coefficients but this is actually a form of a rational function so D is nonlinear. Equation E is nonlinear as the x is squared. So we have

A $x + \pi = e^y$ non linear

B $42x + 67y + 73z = \ln(\pi)$ linear

C $\sin(4)x = y$ linear

D $xy + z = 5$ non linear

E $4\pi = (21x)^2 - \frac{17}{59}y$ non linear

Lets now consider the linear systems of equations

$$42x + 67y + 73z = \ln(\pi)$$

$$\sin(4)x = y$$

$$x - 2y + 6 - 24z = 0$$

We are going to solve this system by manipulating the equations, but before we do, let's look at the set of equations in a form that we might put use as the inputs to a vector form to solve this system using matrices. In the row operations

lecture notes I link the idea of solving systems of equations and matrix forms.

$$42x + 67y + 73z = \ln(\pi)$$

$$\sin(4)x - y + 0z = 0$$

$$-x + 2y + 24z = 6$$

This form would nicely be input into vector form and will be used later in this paper as an example of linear combinations, but for now let us start by isolating one of the variables in one of the equations. We'll start with the easiest and look at the second equation.

$$\sin(4)x - y + 0z = 0$$

which has

$$\sin(4)x = y$$

Plugging this into the third equation we have

$$-x + 2(\sin(4)x) + 24z = 6$$

$$(2\sin(4) - 1)x + 24z = 6$$

$$(2\sin(4) - 1)x - 6 = -24z$$

$$\frac{(2\sin(4) - 1)x - 6}{-24} = z$$

Plugging this into the first equation

$$\begin{aligned}
42x + 67(\sin(4)x) + 73 \frac{(2\sin(4) - 1)x - 6}{-24} &= \ln(\pi) \\
42x + 67(\sin(4)x) - 73 \frac{(2\sin(4) - 1)x - 6}{24} &= \ln(\pi) \\
42x + 67(\sin(4)x) - 73 \frac{2\sin(4) - 1}{24}x + 73 \frac{6}{24} &= \ln(\pi) \\
(42 + 67\sin(4) - 73 \frac{2\sin(4) - 1}{24})x &= \ln(\pi) - 73 \frac{6}{24} \\
(42 + 67\sin(4) - \frac{73\sin(4)}{12} + \frac{73}{24})x &= \ln(\pi) - \frac{73}{4} \\
((67 - \frac{73}{12})\sin(4) + \frac{1081}{24})x &= \ln(\pi) - \frac{73}{4} \\
(\frac{731}{12}\sin(4) + \frac{1081}{24})x &= \ln(\pi) - \frac{73}{4}
\end{aligned}$$

So we have x .

$$\begin{aligned}
x &= \frac{\ln(\pi) - \frac{73}{4}}{\frac{731}{12}\sin(4) + \frac{1081}{24}} \\
&= \frac{24\ln(\pi) - 438}{1462\sin(4) + 1081}
\end{aligned}$$

We can now plug into equations 2 and 3 for y and z . Lets start with equation 3.

$$\begin{aligned}
z &= \frac{(2\sin(4) - 1)x - 6}{-24} \\
&= \frac{(2\sin(4) - 1) \frac{24\ln(\pi) - 438}{1462\sin(4) + 1081} - 6}{-24} \\
&= \frac{(2\sin(4) - 1)(24\ln(\pi) - 438) - 6(1462\sin(4) + 1081)}{-24(1462\sin(4) + 1081)} \\
z &= -\frac{48\sin(4)\ln(\pi) - 24\ln(\pi) - 9648\sin(4) - 6048}{35088\sin(4) + 25944}
\end{aligned}$$

Finally lets calculate y .

$$\begin{aligned}
y &= \sin(4)x \\
&= \sin(4) \frac{24\ln(\pi) - 438}{1462\sin(4) + 1081}
\end{aligned}$$

To conclude lets put this system into the form for a linear combination. So looking at

$$42x + 67y + 73z = \ln(\pi)$$

$$\sin(4)x - y + 0z = 0$$

$$-x + 2y + 24z = 6$$

we can write

$$\begin{bmatrix} 42 \\ \sin(4) \\ -1 \end{bmatrix} x + \begin{bmatrix} 67 \\ -1 \\ 2 \end{bmatrix} y + \begin{bmatrix} 73 \\ 0 \\ 24 \end{bmatrix} z = \begin{bmatrix} \ln(\pi) \\ 0 \\ 6 \end{bmatrix}$$

Where x, y, z are scaling the respective vectors they are attached to.

Notice that the system of equations written as a linear combination has 3×1 matrices ordering the system. In the matrix operations notes we will consider this system again and solve it using the concepts of linear algebra.

Now to calculate weighted grades. Let's consider the example:

In Class Activities	10
Homeworks	25
Quizzes	15
Midterm	20
Final	30

Now we are going to look at linear combinations where grades on assignments are the vectors and this grading system is the scalars.

So first lets turn our scalar weights from percents to decimal form.

so we have

In Class Activities	0.1
Homeworks	0.25
Quizzes	0.15
Midterm	0.2
Final	0.3

This gives the linear combinations of total grade:

$$G_{total} = 0.1G_{Activities} + 0.25G_{Homeworks} + 0.15G_{Quizzes} + 0.2G_{Midterm} + 0.3G_{Final}$$

So if we get the following grades in each category

In Class Activities	100
Homeworks	72
Quizzes	64
Midterm	69
Final	68

we have

$$\begin{aligned}G_{total} &= 0.1(100) + 0.25(72) + 0.15(64) + 0.2(69) + 0.3(68) \\&= 71.8\end{aligned}$$

Now suppose instead we achieved

In Class Activities	98
Homeworks	76
Quizzes	68
Midterm	54
Final	

and we want to know what we need on the final to pass. Then what we do is fill in our information. So

$$70 = 0.1(98) + 0.25(76) + 0.15(68) + 0.2(54) + 0.3G_{Final}$$

And we solve for G_{Final} . Which gives

$$70 = 0.1(98) + 0.25(76) + 0.15(68) + 0.2(54) + 0.3G_{Final}$$

$$70 - (0.1(98) + 0.25(76) + 0.15(68) + 0.2(54)) = 0.3G_{Final}$$

$$20.2 = 0.3G_{Final}$$

$$G_{Final} = \frac{20.2}{.3}$$

$$G_{Final} = 67.3$$

So in order to pass the class we would need a 67.3 on the final.